

EEE554 – Probability and Random Processes

Module 2: Probability Spaces

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From Sets to Probability

In the previous module, we saw that probabilistic questions can be represented as subsets of the sample space.

For example, for a die roll,

$$\Omega = \{1, 2, 3, 4, 5, 6\},$$

the question

“Is the outcome even?”

corresponds to the event

$$A = \{2, 4, 6\}.$$

With this, we can then ask, what is the probability of the event A , $P(A)$?

Thus,

Question $\longrightarrow$ Event $A \subseteq \Omega \longrightarrow$ Probability $P(A)$.
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Which Events Can We Ask About?

Recall that

$$2^{\Omega} = \{A : A \subseteq \Omega\}$$

is the collection of all subsets of Ω . Thus, 2^{Ω} represents all possible events (or questions) about the outcome.

To construct a probabilistic model, we must specify

- 1) which outcomes are possible;
- 2) which events we assign probabilities to;
- 3) what probability is assigned to each event.

Probability Space

These three ingredients define a probability space:

$$(\Omega, \mathcal{F}, P).$$

where Ω is the set of all possible outcomes,

$$\mathcal{F} \subseteq 2^\Omega$$

is the collection of events to which probabilities are assigned, and

$$P : \mathcal{F} \rightarrow [0, 1]$$

assigns a probability to each event.

- Ω : sample space – set of all possible outcomes
- $\omega \in \Omega$: outcome
- $\mathcal{F}$: event space (σ -algebra)
- P : probability measure – assigns a probability value to each event

What Properties Should $\mathcal{F}$ Have?

Suppose $A, B \in \mathcal{F}$. If we can ask

“Is the outcome in A ?”

we should also be able to ask

“Is the outcome NOT in A ?”

and therefore

$$A^c \in \mathcal{F}.$$

Similarly, if we can ask questions about A and B , we should also be able to ask

“Is the outcome in A OR B ?”

and hence

$$A \cup B \in \mathcal{F}.$$

These natural requirements lead to the definition of a σ -algebra.

σ -Algebra

A collection $\mathcal{F} \subseteq 2^\Omega$ is called a σ -algebra if $\Omega \in \mathcal{F}$,

$$A \in \mathcal{F} \implies A^c \in \mathcal{F}, \quad A \cup A^c = \Omega$$

$$A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

In other words, a σ -algebra is a collection of subsets of Ω that contains Ω and is closed under complements and countable unions.

Since


$$A \cap B = (A^c \cup B^c)^c,$$

closure under complements and unions also implies

$$A \cap B \in \mathcal{F}.$$

Consequently,

$$A_1, A_2, \dots \in \mathcal{F} \implies \bigcap_{i=1}^{\infty} A_i \in \mathcal{F},$$

i.e., it is also closed under countable intersections.

Interpretation

A σ -algebra is the collection of probabilistic questions we are allowed to ask.

If $A, B \in \mathcal{F}$, then

$$A^c \in \mathcal{F} \quad (\text{NOT})$$

$$A \cup B \in \mathcal{F} \quad (\text{OR})$$

$$A \cap B \in \mathcal{F} \quad (\text{AND}).$$

Generated σ -Algebras

Suppose

$$\mathcal{A} \subseteq 2^\Omega$$

is a collection of subsets of Ω .

There may be many σ -algebras that contain all the sets in $\mathcal{A}$.

The *σ -algebra generated by $\mathcal{A}$* , denoted by

$$\sigma(\mathcal{A}),$$

is the smallest σ -algebra containing $\mathcal{A}$.

That is,

$$\mathcal{A} \subseteq \sigma(\mathcal{A}),$$

and if $\mathcal{F}$ is any σ -algebra such that $\mathcal{A} \subseteq \mathcal{F}$, then

$$\boxed{\sigma(\mathcal{A}) \subseteq \mathcal{F}.}$$

Thus, $\sigma(\mathcal{A})$ contains exactly the sets that must be included in order to satisfy the properties of a σ -algebra.

Example

Let

$$A \subseteq \Omega.$$

What is the smallest σ -algebra containing A ?

Since a σ -algebra must contain complements,

$$A^c$$

must also be included.

It must also contain

$$\emptyset \quad \text{and} \quad \Omega.$$

$$\Omega = \{1, 2, \dots, 6\}$$

$$A = \{\{2, 4, 6\}\}$$

$$\sigma(A)$$

$$= \{\{2, 4, 6\}, \{1, 3, 5\}\}$$

$$\Omega, \emptyset$$

Therefore,

$$\sigma(A) = \{\emptyset, A, A^c, \Omega\}.$$

Thus, a single nontrivial event A generates a σ -algebra containing four events.

Example

Let

$$\Omega = \{1, 2, 3, 4\},$$

$\sigma(A, B)$

and suppose

$$A = \{1, 2\}, \quad B = \{2, 3\}.$$

Any σ -algebra containing A and B must also contain

$$A^c, \quad B^c,$$

$= \{3, 4\} \quad = \{1, 2\}$

as well as unions and intersections such as

$$A \cap B = \{2\}.$$

From these operations, we can isolate every individual outcome:

$$A \cap (A \cap B)^c = \{1\}, \quad \{2\}, \quad \{3\}, \quad \{4\}.$$

Therefore, every subset of Ω can be constructed, and

$$\sigma(A, B) = 2^\Omega.$$



$$\underline{\text{Ex}} \quad \Omega = \{1, \dots, 6\}$$

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

$$\sigma(A, B) = \left\{ \begin{array}{l} \{1, 2\}, \{5, 6\}, \{3, 4\}, \{1, 2, 3, 4\}, \\ \{1, 2, 5, 6\}, \{3, 4, 5, 6\}, \Omega, \emptyset \end{array} \right\}$$

Kolmogorov's Axioms

We have specified the collection of events $\mathcal{F}$. Now we need to specify how probabilities are assigned to these events. What properties should a probability measure satisfy?

Kolmogorov's axioms specify the fundamental properties that every probability measure must satisfy.

A probability measure

$$P : \mathcal{F} \rightarrow [0, 1]$$

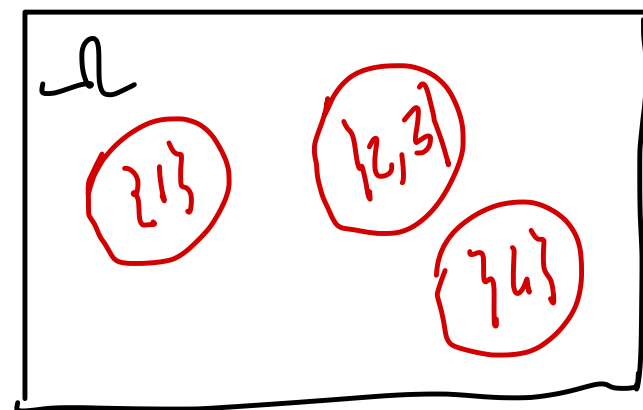
satisfies

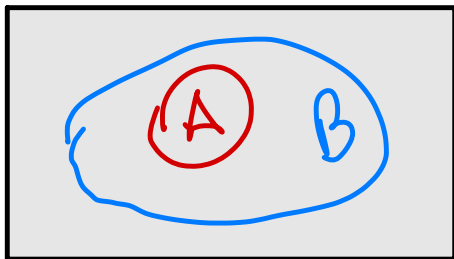
$$P(A) \geq 0, \quad P(\Omega) = 1,$$

and

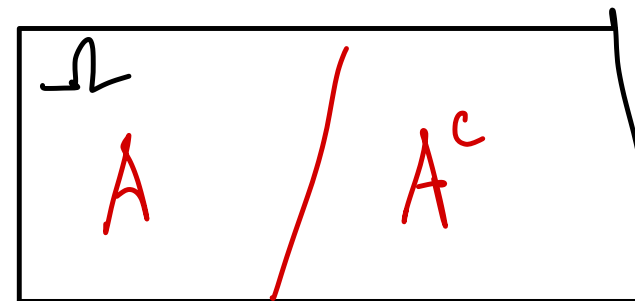
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i),$$

whenever $A_i \cap A_j = \emptyset$, $i \neq j$.





Useful Properties



$$P(\emptyset) = 0, \quad P(A^c) = 1 - P(A),$$

$$1 = P(\Omega) = P(A) + P(A^c)$$

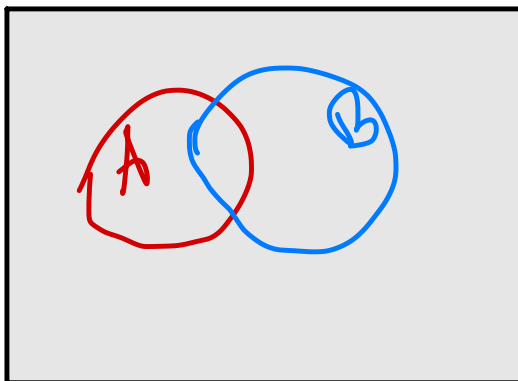
$$A \subseteq B \implies P(A) \leq P(B),$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(A \cap C) - P(B \cap C)$$

$$+ P(A \cap B \cap C).$$



Example: Toss of a Fair Coin

$$\Omega = \{H, T\},$$

$$\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\},$$

$$P(H) = P(T) = \frac{1}{2}.$$

$$P(\{H\} \cup \{T\}) \\ = P(H) + P(T) = 1$$

Hence

$$P(\Omega) = 1, \quad P(\emptyset) = 0.$$

Example: Fair Die

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Since $|\Omega| = 6$, there are $2^6 = 64$ possible events.

Let $A = \{1, 3, 5\}$, $B = \{3, 6\}$. Then

$$A \cap B = \{3\},$$

$$A \cup B = \{1, 3, 5, 6\}.$$

$P(\{1\}) = \dots, P(\{6\}) = 1/6$, hence $P(A \cap B) = P(\{3\}) = 1/6$

$$\begin{aligned} P(A \cup B) &= P(\{1\} \cup \{3\} \cup \{5\} \cup \{6\}) \\ &= P(\{1\}) + P(\{3\}) + P(\{5\}) + P(\{6\}) = 2/3. \end{aligned}$$

Discrete Sample Spaces

Examples

$$\Omega = \{0, 1\}, \text{ (flip of a coin)}$$

$$\Omega = \{1, \dots, 6\}, \text{ (die toss)}$$

$$\Omega = \{0, 1, 2, \dots\}, \text{ (electrons crossing a section of a wire in 1 sec)}$$

Finite or countably infinite.

Continuous Sample Spaces

Examples

$[a, b]$, (temperature in a room)

$\mathbb{R}$, (voltage measured at one instant in a circuit)

$\mathbb{R}^n$, (received signal vector)

Cannot be listed element by element.

Continuous Event Spaces

For $\Omega = \mathbb{R}$, the power set

$$2^{\mathbb{R}}$$

is too large for probability theory.

Instead, we use the

$$\boxed{\mathcal{B}(\mathbb{R})}$$

called the Borel σ -algebra.

It is the smallest σ -algebra containing all open intervals.

- contains every open interval (a, b) ;
- is closed under complements and countable unions.

Equivalently, it is obtained by repeatedly applying complements and countable unions (and hence countable intersections) to open intervals.

Engineering interpretation: In engineering applications, probabilistic questions involving continuous-valued outcomes are almost always formulated in terms of intervals (or combinations of intervals), e.g., “Is $X \in (a, b)$?”, “Is $X > a$?”, or “Is $X \in A$?”, where A is built from intervals. The Borel σ -algebra contains essentially every such event.

Remark.

The Borel σ -algebra does not contain every subset of $\mathbb{R}$. The existence of non-Borel sets is a deep mathematical result (e.g., the Vitali set) that relies on the Axiom of Choice. Such sets practically never arise in engineering applications.

How do we specify the probability measure P in practice?

Discrete spaces: Probability Mass Functions

For a discrete sample space,

$$\forall \omega \in \Omega$$
$$p(\omega) \geq 0,$$

$$\sum_{\omega \in \Omega} p(\omega) = 1.$$

For every event $A \subseteq \Omega$,

$$P(A) = \sum_{\omega \in A} p(\omega).$$

Die roll: $P(\{1, 2, 3\}) = P(\{1\} \cup \{2\} \cup \{3\})$
 $= P(1) + P(2) + P(3)$

Probability Density Functions

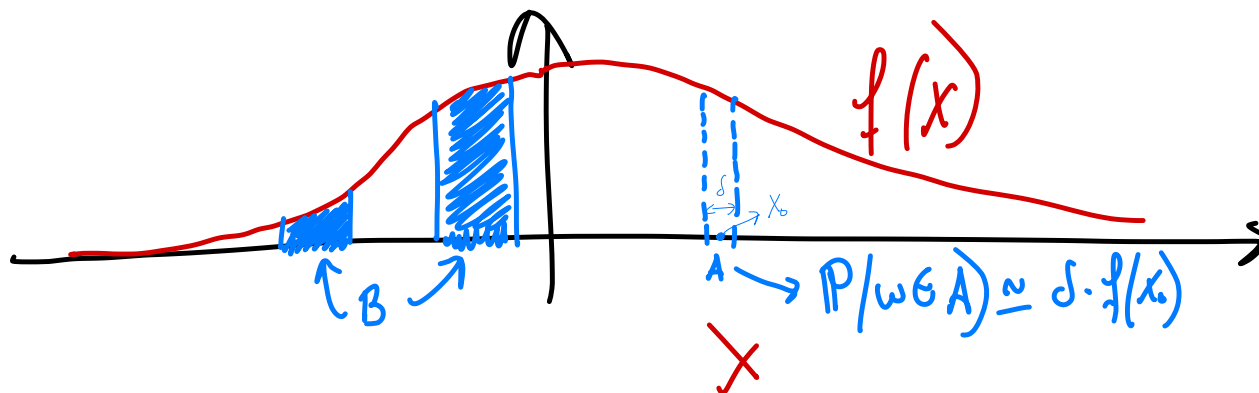
For continuous sample spaces, we define the *density function*

$$f(x) \geq 0,$$

$$\int_{\mathbb{R}} f(x) dx = 1.$$

Probability is computed using

$$P(A) = \int_A f(x) dx.$$



Cartesian Products

Repeated experiments are modeled by the Cartesian product

$$\Omega^k = \underbrace{\Omega \times \Omega \times \cdots \times \Omega}_{k \text{ times}}.$$

An outcome is

$$(\omega_1, \omega_2, \dots, \omega_k),$$

where $\omega_i \in \Omega$ is the outcome of the i th experiment.

Example: a die is rolled k times.

$$\Omega = \{1, 2, 3, 4, 5, 6\},$$

$$\Omega^k = \{(\omega_1, \dots, \omega_k) : \omega_i \in \Omega\}.$$

Sometimes the number of repetitions is not known in advance.

In this case, the sample space is

$$\Omega^\infty = \Omega \times \Omega \times \cdots ,$$

whose outcomes are infinite sequences

$$(\omega_1, \omega_2, \omega_3, \dots).$$

Example: Repeated Binary Trials

Suppose $\Omega = \{0, 1\}$, where

$$1 = \text{success}, \quad 0 = \text{failure}.$$

The experiment is repeated until the first success occurs.

Consider the event

$$E = \{\text{an even number of failures before the first success}\}.$$

Then

$$E = \{1, 001, 00001, 0000001, \dots\}.$$

Equivalently,

$$E = \{0^{2^n}1 : n = 0, 1, 2, \dots\}.$$

Remark.

Although every sequence in E is finite and ends with a 1, its length is not bounded.

Indeed,

$$E = \{0^{2^n}1 : n = 0, 1, 2, \dots\},$$

so the first success may occur arbitrarily late.

Example:

$$p(0) = p(1) = \frac{1}{2}.$$

Then

$$P(E) = \sum_{n=0}^{\infty} P(\text{2n failures followed by 1 success})$$
$$P(E) = P(1) + P(001) + P(00001) + \dots$$

We will later show that, under independent binary trials,

$$P(001) = \left(\frac{1}{2}\right)^3,$$

and more generally,

$$P(0^{2n}1) = \left(\frac{1}{2}\right)^{2n+1}.$$

$$\text{Geom. sum: } \sum_{k=0}^{\infty} a^k = \frac{1}{1-a}, \quad |a| < 1$$

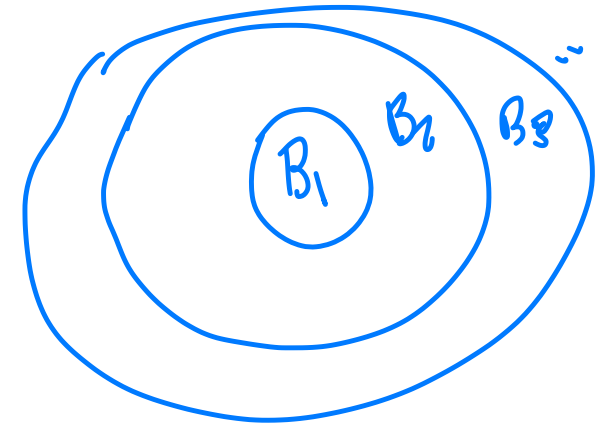
Then we conclude:

$$\begin{aligned} P(E) &= \frac{1}{2} + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^5 + \dots \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2n+1} \\ &= \frac{2}{3}. \end{aligned}$$

Further properties: Continuity of Probability

If $B_1 \subseteq B_2 \subseteq \dots$, then

$$\lim_{n \rightarrow \infty} P(B_n) \Rightarrow P\left(\bigcup_{n=1}^{\infty} B_n\right).$$



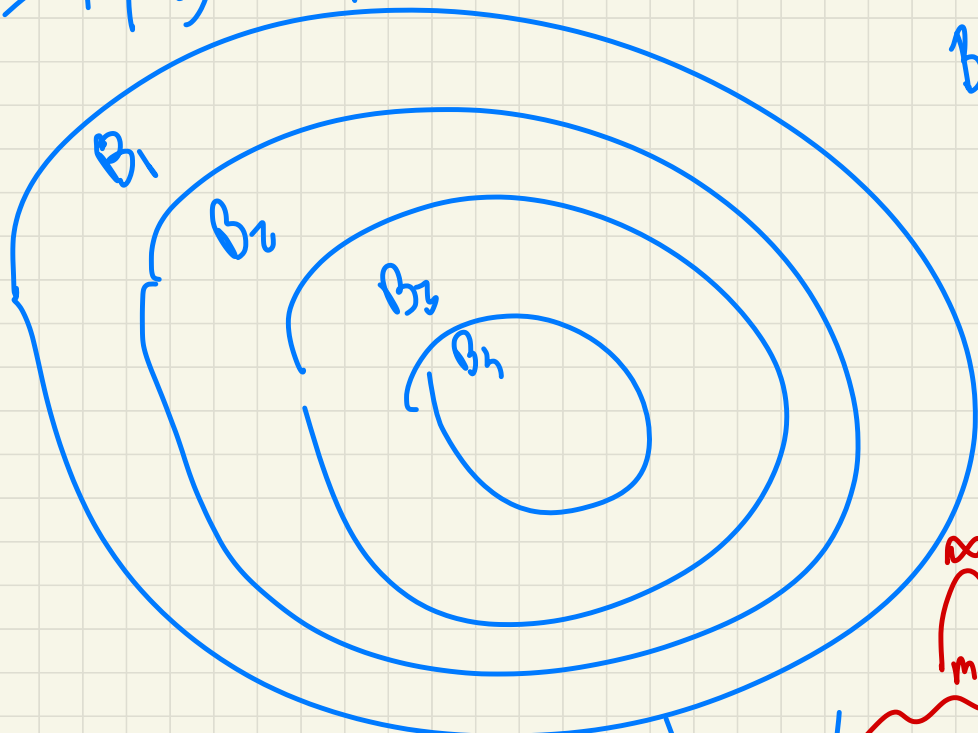
If $B_1 \supseteq B_2 \supseteq \dots$, then

$$\lim_{n \rightarrow \infty} P(B_n) \Rightarrow P\left(\bigcap_{n=1}^{\infty} B_n\right).$$

$$B_n = \bigcap_{m=1}^n B_m$$

$$\lim_{n \rightarrow \infty} P(B_n) = \lim_{n \rightarrow \infty} P\left(\bigcup_{m=1}^n B_m\right)$$

$$P(B_1) \supseteq P(B_2) \supseteq P(B_3) \dots$$



$$\begin{aligned} B_2 &= B_2 \cap B_1 \\ B_3 &= B_3 \cap B_2 \cap B_1 \\ &\vdots \\ B_n &= B_n \cap \dots \cap B_1 \end{aligned}$$

$$\lim_{n \rightarrow \infty}$$

$$P(B_n) = \lim_{n \rightarrow \infty}$$

$$P\left(\bigcap_{m=1}^n B_m\right)$$

$$= P\left(\lim_{n \rightarrow \infty} \bigcap_{m=1}^n B_m\right)$$

$$\bigcap_{m=1}^{\infty} B_m$$